Beyond the Algorithm: Diagnosing the Dominance of User Agency in the Formation of Digital Echo Chambers

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Introduction

It is widely acknowledged that social media has become a primary arena for political discussion and information exchange, raising widespread concern about how individuals encounter and interpret digital content. One of the most pressing issues in this domain is the observation that citizens are increasingly trapped in echo chambers — insulated digital environments where users encounter only information that reinforces their pre-existing beliefs, effectively silencing disagreement and weakening the conditions necessary for informed democratic citizenship (Ahmmad et al., 2025, p. 2).

While algorithms are often presented as the main drivers of this ideological isolation, recent literature such as Parmelee & Roman (2020) (p. 2) and Ahmmad et al. (2025) (p. 2) suggests that user behavior, specifically the psychological drive to seek affirmation and avoid dissonance, may be an equally, if not more, potent architect of isolation. This presents a key unresolved tension in the current literature: whether echo chambers stem primarily from technological architecture or from predictable cognitive habits. While algorithmic curation is well-documented, the interaction between user agency and these systems remains insufficiently evaluated.

This gap in our understanding is critical. If we assume algorithms are the sole culprits, our solutions will be limited to platform regulation. However, if behavioral tendencies of users substantially sustain echo chambers, such interventions will fail to address the underlying drivers of cognitive isolation. Therefore, it is essential to isolate and diagnose the role of the user — a variable that is less understood yet potentially more decisive than the algorithm itself.

To gain a better understanding of this dynamic, this study attempts to examine the degree to which selective exposure, a psychological phenomenon where users actively seek out information that confirms their worldviews, contributes to echo-chamber formation beyond algorithmic curation. Drawing on empirical findings from eight post-2020 peer-reviewed articles, the analysis focuses on how platform design intersects with psychological tendencies such as confirmation bias and avoiding critical thought. Importantly, the post-2020 period was deliberately chosen because it coincides with heightened political polarization due to the 2020 U.S. election and widespread remote social-media use during the COVID-19 pandemic (Li et al., 2025, p. 12). It seems plausible that these contextual factors have intensified the development of echo chambers, making post-2020 research particularly relevant for diagnosing these user-driven behaviors in their most acute form.

Clarifying this relationship is of great importance for effective policy design. If behavioral tendencies of users substantially sustain echo chambers, interventions limited to algorithmic transparency or platform regulation will not fully address the underlying drivers of cognitive isolation. Consequently, strategies that support media literacy and healthier digital habits will become equally necessary.

This leads us to question:

RQ: *To what extent does user-driven selective exposure contribute to the formation of social media echo chambers, relative to algorithmic curation?*

In this study, we aim to employ a comparative synthesis of eight empirical studies to build a cohesive argument. Rather than listing findings in isolation, we orchestrate a dialogue between scholars, contrasting their methodologies and findings to reveal the nuanced interplay between human agency and algorithms. Based on this synthesis, we hypothesize that user behavior is the central mechanism that sustains and amplifies cognitive isolation, with algorithms serving as catalysts rather than root causes.

The remainder of the paper is organized as follows. First, we examine the contrasting user-driven architectures (Reddit) with algorithm-driven ones (TikTok). This comparative approach is essential to isolate the variable of user choice, demonstrating that granting users more agency can, counterintuitively, lead to deeper isolation. Next, having established that user choice can drive polarization, we turn to diagnose the specific cognitive mechanisms that fuel this self-isolation. This psychological foundation is necessary to understand the subsequent section on "Structural Ambivalence," where we analyze how specific platform designs interact with these cognitive biases to either amplify or fracture echo chambers. Finally, we conclude with implications for policy and literacy, arguing that effective interventions must target the root cognitive drivers of isolation rather than merely regulating algorithmic outputs. Through reframing echo chambers as products of both technological and psychological forces, this study seeks to advance a more balanced understanding of responsibility in the digital mediascape and inform future research and policy.

The Platform Paradox

A central tension in the literature is the assumption that if algorithms are the primary drivers of echo chambers, then platforms that offer users more agency and control should theoretically produce more diverse information diets. If users are free to choose, the logic goes, they will choose a balanced diet of information. However, a comparative analysis of empirical data reveals a counterintuitive reality, which we term the "Platform Paradox". In other words, platforms designed around user choice often produce higher levels of polarization than those driven by opaque algorithms. In the following section, we analyze how this phenomenon makes itself visible across different platforms and influences user agency.

To illustrate, Kitchens et al. (2020) arrived at this counterintuitive finding through a rigorous study of desktop browsing sessions from 2012 to 2016. By tracking the specific click-paths of users, measuring where they came from (referrals) and where they went (news sites), they could observe behavior in the wild. This methodology allowed them to bypass the self-reporting bias often found in survey data (p. 1629). Their analysis yielded a striking contrast showing that information presented from Facebook's feed showed greater information diversity. By contrast, "Reddit usage was associated with a significant decrease in diversity and an increase in partisan slant" (p. 1633).

This finding is critical because of the architectural differences between the two platforms. Facebook users are surrounded by an environment almost out of their control, with an opaque algorithm that self-selects information to be presented to the user. Conversely, Reddit's algorithm runs on "subreddits" which are communities that users strictly self-select to join and consume information from (Reddit doesn't automatically present information, it can only suggest subreddits to guide users towards specific communities). Essentially, the user builds their own echo chamber, brick by brick. The fact that the platform driven by user-self-selection (Reddit) produced higher polarization suggests that users are more efficient at isolating themselves than an algorithm's attempt to isolate them.

However, when we shift to different mediums which also revolve around an opaque framework (such as TikTok), we observe a different dynamic. In contrast to Reddit's active community selection, TikTok's "For You" feed is driven by a sophisticated recommendation engine that minimizes reliance on pre-existing social networks. Li et al. (2025) found that TikTok's distinct political echo chambers are reinforced by "positive social reinforcement" (likes and views) (p. 17). This creates a self-reinforcing loop, where the most liked content is given more priority by the algorithm, presenting itself to more users (regardless of being factually right) and amplifying the boundaries of the echo chamber.

In addition to the above, their analysis of over 16 million videos revealed that users with extreme political views who received positive community engagement were statistically more likely to produce even more polarized content. In this context, the algorithm acts as a mirror that reflects the user's desires back to them with increasing intensity. Unlike algorithmic curation or user self-selection dominating each other, TikTok displays a system where both factors mend together to create these filter bubbles. Thus, even when opaque algorithms are introduced, the paradox defines that beyond just user agency and algorithms, the outcome is heavily dictated by the user's underlying psychological drive to find and engage with content that validates their identity.

False Illusion of User Innocence

If the Platform Paradox establishes that users are the architects of their own isolation, we must next examine the psychological blueprints they use to build these chambers of isolation. The literature synthesis in this section demonstrates that user behavior contributes to echo chamber formation not merely through a passive preference for agreement (selective exposure), but through active selective avoidance. By prioritizing mental comfort over critical analysis, users deliberately filter their environments to exclude dissonance.

Focusing on results from behavioral studies, results further confirm that users are not passive victims of curation but active agents of avoidance. Parmelee and Roman (2020) challenged the passive-user narrative through a survey of 309 politically active Instagram users, explicitly recruiting those who followed political leaders (p. 5). Their findings were stark, showing that on Instagram, 79.9% of politically active users follow only leaders they agree with while only 15.3% follow opposing views (p. 5).

Importantly, Parmelee and Roman (2020) found that "ideological strength" predicted not just a preference for agreement, but an active avoidance of disagreement (Parmelee & Roman, 2020, p. 8). This suggests that users are not merely drifting into echo chambers but actively chiseling their feeds to remove conflicting voices. This user-driven filtering creates an asymmetric echo chamber effect, where conservatives were found to be significantly more likely to isolate themselves than liberal counterparts, hinting towards users attempting to place themselves in familiar situations varying with political ideology.

Consistent with these findings, Rhodes (2022) provides a cognitive explanation for this behavior through a survey experiment with 1,749 participants, randomly assigning them to simulated news feeds containing either homogenous (agreeable) or heterogeneous (mixed) content (p. 10-11). This experimental design allowed Rhodes to isolate the specific effect of the environment on the user's critical thinking skills. He found that a primary mechanism sustaining these bubbles was cognitive. Participants in simulated filter bubbles engaged in higher levels of “heuristic processing” (a mental shortcut favoring quick, affect-based judgments over deep analysis), abandoning critical thought because the environment felt safe and familiar (Rhodes, 2022, p. 7).

This is not something unfamiliar as confirmation bias from like-minded individuals amplifies an opinion (whether it be wrong), driving more people to believe it as it is presented repeatedly, but each time more confidently. Rhodes emphasizes this with his finding that exposure to congenial information “may drive the effects of fake news more than other variables like age or partisanship” (p. 11). Consequently, the echo chamber is solidified by the user’s cognitive laziness to challenge a presented opinion. When users curate their feeds to exclude dissonance, they create an environment that degrades their own ability to process truth, making them susceptible to fake news not because the algorithm hid the truth, but because they chose not to see it.

Structural Ambivalence

While users drive the selection, the architectural design of social media platforms dictates the accelerant potential of echo chamber formation. This section examines the structural features of digital platforms to demonstrate how different algorithms actively construct the conditions for isolation. The literature reveals that digital platforms possess what Palmieri (2024) calls "ambivalent potential" (p. 2602). This theoretical framework presents that platforms are not inherently isolating but rather, depending on their specific mechanisms (such as algorithmic boosting or open interaction) they can either restrict communication to a closed loop of reaffirming ideas or expose users to a variety of voices, forcing them to think beyond their biases. The following studies illustrate this dichotomy, showing how specific structural mechanisms can either accelerate polarization or dissect it.

On the one hand, Ferraz de Arruda et al. (2024) illustrated the accelerant potential using computational modelling. In their methodology, they used agent-based modeling, a type of computer simulation where virtual 'agents' interact according to set rules. This allowed them to tweak variables impossible to control in real life, such as introducing 'stubborn' users who never change their minds (to represent polarizing but charismatic public figures) (p. 4). They demonstrated that “priority users” (those with algorithmic boosting, similar to a verified checkmark on social media platforms) can catalyze polarization, and these are often people in positions of power or influential figures (who have a strong voice and an audience who are willing to listen to them) (p. 9). Their simulations showed that these figures could induce a transition from consensus to polarization and this was amplified when these individuals were given priority status (meaning that followers would more frequently encounter their opinions and socially enforce them through engagement). This shows that the power to manipulate echo chamber formation and information environments often lies within the hands of a minority of individuals and this access is handed to them through the platform's superiority features.

On the other hand, the structure is ambivalent across different platforms and the amplification is not universal. Sun et al. (2022) found only a "weak echo chamber effect" on Youtube regarding vaccine misinformation (p. 10), through a social network analysis of thousands of comment-reply structures on a viral Youtube video regarding "vaccine misinformation" (p. 6). Upon data extraction, their study found that "intergroup interaction" (users commenting on opposing videos) diminished the effect of the echo chamber, a contention to Ferraz de Arruda et al.'s findings (p. 10). In contrast to the "feed" based systems of Tiktok, Instagram or Facebook, Youtube's structure allowed for users from opposing viewpoints to collide in the comments, which would amplify them, disrupting the isolation present in the echo chamber. This pattern of results is consistent with Palmieri's (2024) theoretical result that social media has an "ambivalent potential": it expands chances for homogeneity just as much as it allows users to reduce it. Thus, platforms that prioritize broadcast dynamics and elite voices (like X/Twitter) tend to harden echo chambers. Conversely, platforms that allow for cross-pollination of users through open interaction (like YouTube comments) can fracture these bubbles, even if the discourse is contentious. The trap of the echo chamber is not inevitable; it is a design choice and it is effective only because the user walks into it willingly.

Conclusion

The purpose of this study was to gain a better understanding of the interplay between algorithms and user agency in digital isolation. Overall, the results of this research provide supporting evidence that apart from algorithm curation, users play a major role in the formation of echo chambers in social media. Some platforms become polarized when users self-select the content they consume (as on Reddit), while others become polarized because the algorithm pushes engaging or extreme content (as on TikTok).

Furthermore, behavioral studies confirm that users actively filter out dissonance to protect their own cognitive comfort, using the algorithm to confirm their biases. Side by side, algorithms serve as a catalyst to either amplify or diminish the effect of echo chambers that users generated. However, this agency is mediated by platform design: structural features such as priority users can enhance polarization, while platforms that allow intergroup interaction such as YouTube comment section diminish these bubbles In short, echo chambers persist because both platform and user contribute in narrowing the information scope.

While there has been a lot of study regarding how algorithms form echo chambers in social media, the role of users in this phenomenon remains underexplored. Existing research mentions that users indeed contribute to the phenomenon, however the bigger picture is still fragmented. The original contribution of this review is the crystallization of a feedback loop between users and algorithms: algorithms do not simply imprison users but users actively build the prison, and algorithms merely reinforce the bars. Beyond offering a more nuanced framework that includes both user agency and platform structure, this study also has practical importance. The results strongly imply that the government should be informed in policy design regarding platform regulations. Educational strategies or policies must pivot away from simply technical media literacy to cognitive literacy:

through teaching users to recognize their own heuristic processing and become self-aware of cognitive laziness. If we are to escape echo chambers, we must learn to navigate and acknowledge all the diverse voices that exist around us.

Limitations of Existing Research

Although the present results clearly support the hypothesis, it is appropriate to recognize several potential limitations. First, as noted by Ahmmad et al. (2025) there is a “marked geographical imbalance” (p. 14) as the primary empirical evidence, specifically the user browsing history analyzed by Kitchens et al. (2020), is skewed heavily toward the Western social context (p 12). This reliance results in a significant lack of global generalizability, as the polarization dynamics inherent to the United States' two-party system differ fundamentally from the discourse of non multi-party democracies. Furthermore, the specific regulatory environment of the U.S., characterized by minimal state intervention, contrasts sharply with the stricter content moderation found in the European Union or the state-controlled internet infrastructures of authoritarian regimes.

Secondly, despite our attempts to isolate algorithms and user-selective exposure, real-world scenarios involve them working mutually reinforcingly which creates an unrealistic testing environment. Because users react to algorithmic suggestions and algorithms react to user-behaviour, establishing a clear and direct causal link is almost impossible. The limitation resulting from this is that it is difficult to quantify the effect size of each domain, since each mechanism constantly feeds off the other. Kitchens et al. (2020) explicitly acknowledges in their own research limitations that while they can identify an effect, they “cannot distinguish whether the changes are driven by algorithmic curation (platform choice) or user self-selection (user choice)" (p. 1642). Through asserting user dominance, our research may be overstating the ability to disentangle these forces.

It is also possible that by relying heavily on these quantitative "effect sizes" and behavioral norms in an attempt to oversimplify relationships, the framework ignores external sociopolitical factors which could shape an individual's thoughts beyond social media. Rhodes (2022) notes that asymmetric polarization (where Republicans were less influenced by heterogeneous information streams) may be due to external factors such that "conservative outlets continuously attack mainstream news" (p. 15). This suggests that echo chambers are maintained not just by user psychology but by broader media ecology.

In addition, it is important to emphasize Palmieri's (2024) idea regarding the ambivalent potential of social media in the face of infinite choices. Although the binary outcome that users lack critical thinking and have confirmation bias, resulting in the creation of echo chambers may seem straightforward, individual expression is not as simple. Palmieri (2024) argues that users are navigating a system that ambivalently floods them with new information (increasing possibilities) while forcing them to filter that information to protect their moral identity (decreasing possibilities) (p. 2600). It seems plausible through her argument that social media is a double-edged sword, it offers us infinite choices, but this very abundance forces us to filter aggressively to protect our sanity. Therefore, an echo chamber is not just a user's failure, but a necessary way users cope with the overwhelming range of choices in the digital space.

Scope for Future Research

Accompanying the broad variety of sources are several lingering gaps that are left unaddressed, which leaves scope for future researchers to expand upon this framework. One of the most pressing issues is the need to bridge the cultural disparity in current literature. While Li et al. (2025) calls for “cross-cultural comparative designs” (p. 18), this must be translated into concrete practice. In practical terms, this entails adopting experimental designs that explicitly replicate these studies in diverse political environments to understand how strong these effects remain across different geography and demography. For example, researchers should investigate whether the dominance of user agency persists in multi-party democracies like Germany or Brazil, or if it manifests differently in authoritarian contexts like China, where state-controlled information environments impose different constraints on user choice. Understanding these nuances is critical to determining if the "Platform Paradox" or user behaviour in relation to the formation of echo chambers is a universal phenomenon or unique to the polarized two-party system of the United States.

In terms of future research, it would also be useful to extend the current findings by moving beyond surveys and establishing observations through field work designs. This would involve tracking real-time user behavior over long periods to isolate variables more precisely, allowing us to disentangle the roles of algorithmic curation versus user self-selection.

Taken together, understanding the user’s role is the only path toward developing effective interventions, since algorithms lack the incentive to reorganize their framework (since businesses hoard revenue of this exact phenomenon). If the echo chamber is indeed a prison constructed by user agency and reinforced by algorithmic architecture, then true digital liberty does not necessarily come from regulating the machine. True liberty arrives not by avoiding the friction of democratic discourse, but by teaching the user how to navigate it.

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